
Investments Final Project

Spring 2026

Stock market reactions to FOMC rate decisions

[Your Name]

Department of Finance
National Taiwan University

June 2026

JEL classification: E43, E44, E58, G12, G14

Keywords: event study, FOMC, monetary policy, Fama–French model, market efficiency, abnormal returns

Stock market reactions to FOMC rate decisions¹

[Your Name]

June 2026

Abstract

I run an event study of nine FOMC rate-change announcements between March 2022 and December 2025 to ask whether US equity prices behave around these announcements in a way consistent with semi-strong-form market efficiency. Expected returns come from the Fama–French three-factor (FF3) model, estimated separately for each (firm, event) pair on a $[-250, -30]$ trading-day pre-event window. The sample is 30 large-cap US firms across seven sectors, giving 270 firm-event observations. The pooled cumulative average abnormal return (CAAR) reaches roughly -0.85% by the announcement day and then sits there. None of the three test windows — $[-1, +1]$, $[0, +1]$, $[0, +5]$ — rejects the null of zero abnormal return, with p -values of 0.60, 0.27 and 0.99. Most of the cumulative drift takes place between $t = -5$ and $t = -2$, before the announcement; from $t = -1$ onward the trajectory flattens. The aggregate result is what efficiency predicts and is robust to splitting the sample by direction (hikes vs. cuts) and to an event-clustered bootstrap. The sector-level numbers are more informative. Banks gain a statistically significant $+1.23\%$ in the $[-1, +1]$ window ($t = +3.41$, $p = 0.001$) and utilities lose a statistically significant -0.83% ($t = -3.28$, $p = 0.002$), with the remaining sectors falling in between. A market-wide null does not mean that nothing happens around an FOMC announcement; it means that the sector responses cancel.

JEL classification: E43, E44, E58, G12, G14.

Keywords: event study, FOMC, monetary policy, Fama–French model, market efficiency, abnormal returns.

¹ Final project submitted in partial fulfilment of the Investments course requirements, Spring 2026. The analysis uses publicly available data from Yahoo Finance and the Fama–French Data Library maintained by Kenneth R. French at Dartmouth Tuck School of Business. All errors are the author’s own.

1 Introduction

The starting point for this paper is a narrow question. When the Federal Open Market Committee announces a change in its target rate, what do US equity prices do in the days around the announcement? Under the semi-strong form of the Efficient Market Hypothesis the answer should be: not much in the days after, since whatever information the statement carries should already be priced in by the close on announcement day. Under the alternative, the post-event window should show continued drift (under-reaction) or a partial reversal (over-reaction). The empirical question is which of these the data look like.

The literature has worked on this for decades. [Bernanke and Kuttner \(2005\)](#) and [Kuttner \(2001\)](#) use Fed-funds-futures-derived surprises to show that the unexpected component of FOMC actions does move equity prices. Whether the average effect represents an exploitable departure from efficiency, or just risk repricing that anyone could in principle have anticipated, remains contested.

The empirical strategy I use is the event-study framework summarised in [MacKinlay \(1997\)](#), with the statistical conventions for significance testing on daily abnormal returns set out in [Brown and Warner \(1985\)](#). Nine rate-change announcements between March 2022 and December 2025 are studied. The window covers the full hiking cycle of 2022 and 2023, the September 2024 pivot, and the easing cadence of late 2024 and 2025. The sample is 30 large-cap US firms grouped into seven sectors selected to span the cross-section of interest-rate sensitivity: six banks, six REITs, five utilities, six mega-cap technology names, three consumer-discretionary names, two industrials, and two energy firms. Abnormal returns are constructed against a Fama–French three-factor benchmark estimated for each (firm, event) pair on its $[-250, -30]$ trading-day pre-event window. The event window is $[-5, +5]$ for the trajectory plot. Three test windows are used for the formal t -tests: $[-1, +1]$ for the immediate reaction, $[0, +1]$ for the announcement day plus one, and $[0, +5]$ for post-announcement drift.

The pooled $\overline{CAR}$ across the 270 firm-event observations is statistically zero in every test window. Most of the cumulative movement is realised before the announcement, between $t = -5$ and $t = -2$, and from $t = -1$ onward the trajectory is flat. The narrowest of the three windows is the noisiest but still does not reject the null. At face value the aggregate reads like semi-strong-form efficiency: by 14:00 ET on FOMC day the news is already in the price.

The aggregate misses the sector-level result. The sign of the sector-mean CAR lines up with what rate-sensitivity theory predicts. Banks and industrials drift up, utilities and energy drift down, and technology and consumer discretionary are flat or close to it. The spread between the most positive and most negative sector means is about 1.5 percentage points in the $[0, +5]$ window. That is an order of magnitude larger than the pooled mean. A market-wide null does not mean that nothing was happening on FOMC days; it means that the sectoral responses average out.

The rest of the paper is organised as follows. Section 2 describes the sample, the events, and the data sources. Section 3 sets out the methodology. Section 4 reports the results. Section 5 discusses and concludes.

2 Data

2.1 Sample of firms

Thirty US-listed, large-capitalisation firms are selected to span the full range of theoretical interest-rate sensitivity, so that the cross-sectional structure of the abnormal-return response can be examined. All firms are continuously listed and liquid throughout the 2021–2026 window. Table 1 lists the sample together with the ex-ante directional hypothesis for each sector following an FOMC rate *hike*.

Table 1: Sample of 30 firms grouped by sector. The final column reports the ex-ante directional hypothesis on the sign of AR following an FOMC rate hike. For rate cuts the predicted signs reverse.

Sector (n)	Firms (ticker)	Hypothesis
Banks (6)	JPMorgan (JPM), Bank of America (BAC), Wells Fargo (WFC), Citigroup (C), Goldman Sachs (GS), Morgan Stanley (MS)	$AR > 0$
REITs (6)	Realty Income (O), Simon Property (SPG), Prologis (PLD), Equinix (EQIX), American Tower (AMT), Welltower (WELL)	$AR < 0$
Utilities (5)	NextEra (NEE), Duke Energy (DUK), Southern (SO), AEP (AEP), Exelon (EXC)	$AR < 0$
Tech (6)	Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), Amazon (AMZN), NVIDIA (NVDA), Meta (META)	$AR < 0$
Consumer Disc. (3)	Home Depot (HD), Nike (NKE), Starbucks (SBUX)	$AR < 0$
Industrials (2)	Caterpillar (CAT), Deere (DE)	$AR \lesseqgtr 0$
Energy (2)	ExxonMobil (XOM), Chevron (CVX)	control

2.2 Events

Nine FOMC target-rate change announcements over March 2022 through December 2025 are studied, spanning the full hiking phase (2022–2023), the September 2024 pivot, and the easing cadence of late 2024 and 2025. Each event is the date on which the FOMC statement is released; abnormal returns are measured on the trading day of release relative to the prior close. Dates are obtained from the Federal Reserve Board’s Open Market Operations history.²

Table 2: FOMC rate-change announcements used as events ($N = 9$).

Date ($t = 0$)	Action	Notes
17 March 2022	+25 bp	First hike of the post-COVID cycle
16 June 2022	+75 bp	First +75 bp move since 1994
22 September 2022	+75 bp	Hawkish surprise; hot prior-week CPI
23 March 2023	+25 bp	Hike during the SVB / regional-bank stress
19 September 2024	−50 bp	First cut of the cycle; market pricing split
19 December 2024	−25 bp	“Hawkish cut” (dot-plot shock)
18 September 2025	−25 bp	Opening cut of the 2025 easing cadence
30 October 2025	−25 bp	Continuation cut
11 December 2025	−25 bp	Final cut before extended pause

2.3 Sources and pre-processing

Daily adjusted close prices come from Yahoo Finance via the `yfinance` Python library over 1 January 2021 through 13 June 2026, which gives 1 367 trading days per ticker. Daily returns are computed as $R_{i,t} = P_{i,t} / P_{i,t-1} - 1$. The Fama–French daily research factors (MKT – RF, SMB, HML) and the risk-free rate RF come from Kenneth French’s Data Library at Dartmouth Tuck School of Business.³ One detail worth flagging: the factor series is published in percent, not decimal, and must be divided by 100 before merging with returns. Forgetting this step is a common cause of nonsensical abnormal-return magnitudes in event-study replications.

² federalreserve.gov/monetarypolicy/openmarket.htm, accessed 13 June 2026.

³ mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html, accessed 13 June 2026.

3 Methodology

3.1 Estimation window

For each pair (firm i , event e), factor loadings are estimated on the trading-day window

$$\tau \in [-250, -30]$$

relative to the event day $t = 0$. This is approximately one calendar year of daily returns, terminating one calendar month before the event so that the estimation window is not contaminated by potential pre-event leakage or drift. The estimation procedure is repeated independently for each of the $30 \times 9 = 270$ firm-event pairs, so loadings are permitted to vary across events.

3.2 Factor model

The Fama–French three-factor model of [Fama and French \(1993\)](#) is estimated by OLS:

$$R_{i,\tau} - \text{RF}_\tau = \alpha_{i,e} + \beta_{i,e}^{\text{MKT}} (\text{MKT} - \text{RF})_\tau + \beta_{i,e}^{\text{SMB}} \text{SMB}_\tau + \beta_{i,e}^{\text{HML}} \text{HML}_\tau + \varepsilon_{i,\tau}. \quad (1)$$

3.3 Abnormal and cumulative abnormal returns

For each day t in the event window $[-5, +5]$ the counterfactual expected return is

$$\widehat{E}[R_{i,t}] = \text{RF}_t + \hat{\alpha}_{i,e} + \hat{\beta}_{i,e}^{\text{MKT}} (\text{MKT} - \text{RF})_t + \hat{\beta}_{i,e}^{\text{SMB}} \text{SMB}_t + \hat{\beta}_{i,e}^{\text{HML}} \text{HML}_t, \quad (2)$$

and the abnormal return and the firm-event cumulative abnormal return follow as

$$AR_{i,t} = R_{i,t} - \widehat{E}[R_{i,t}], \quad \text{CAR}_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}. \quad (3)$$

3.4 Pooled averages

Pooling across all $N = 270$ firm-event observations yields the average abnormal return at each event-time and its cumulative analogue:

$$\overline{AR}_t = \frac{1}{N} \sum_{i=1}^N AR_{i,t}, \quad \overline{\text{CAR}}(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} \overline{AR}_t. \quad (4)$$

3.5 Significance tests

For each test window $[\tau_1, \tau_2]$ a one-sample t -test of the cross-sectional distribution of $\text{CAR}_i(\tau_1, \tau_2)$ against the null of zero abnormal return is computed:

$$t = \frac{\overline{\text{CAR}}(\tau_1, \tau_2)}{\hat{\sigma}_{\text{CAR}} / \sqrt{N}}, \quad \hat{\sigma}_{\text{CAR}} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\text{CAR}_i - \overline{\text{CAR}})^2}. \quad (5)$$

With $N = 270$ the critical value at $\alpha = 5\%$ is $|t| \approx 1.97$. Three windows are reported: $[-1, +1]$ for the immediate reaction, $[0, +1]$ for the announcement day plus one, and $[0, +5]$ for the post-announcement drift. A Bonferroni correction across the three windows would set the per-test level at 1.67%; the qualitative conclusions below do not depend on which level is used.

3.6 Why FF3 rather than FF5 or Carhart

The FF3 model is preferred to the Fama–French five-factor (FF5) and Carhart four-factor alternatives for two reasons. The additional factors (profitability and investment in FF5; momentum in Carhart) have no clear theoretical link to interest-rate-policy news; they would absorb cross-sectional return variation that is conceptually unrelated to the event. The momentum factor in particular is constructed from past returns and therefore risks absorbing genuine Fed-induced effects rather than removing benchmark exposure. FF3 is also the workhorse model in event-study applications, which makes the results easier to compare across studies.

3.7 Robustness checks

Three robustness checks are performed in addition to the pooled tests. The sample is split by direction (hikes vs. cuts) and the t -tests are repeated on each subsample, since the theoretical abnormal-return signs are opposite for the two directions. Sector-level one-sample t -tests against zero CAR are reported for the same three windows to determine which sectors drive the cross-sectional pattern. The $[0, +5]$ headline window is also re-tested using a bootstrap that resamples whole events with replacement (2,000 replications). Resampling at the event level preserves the within-event cross-sectional correlation among the 30 firms that the naive t -test ignores.

4 Results

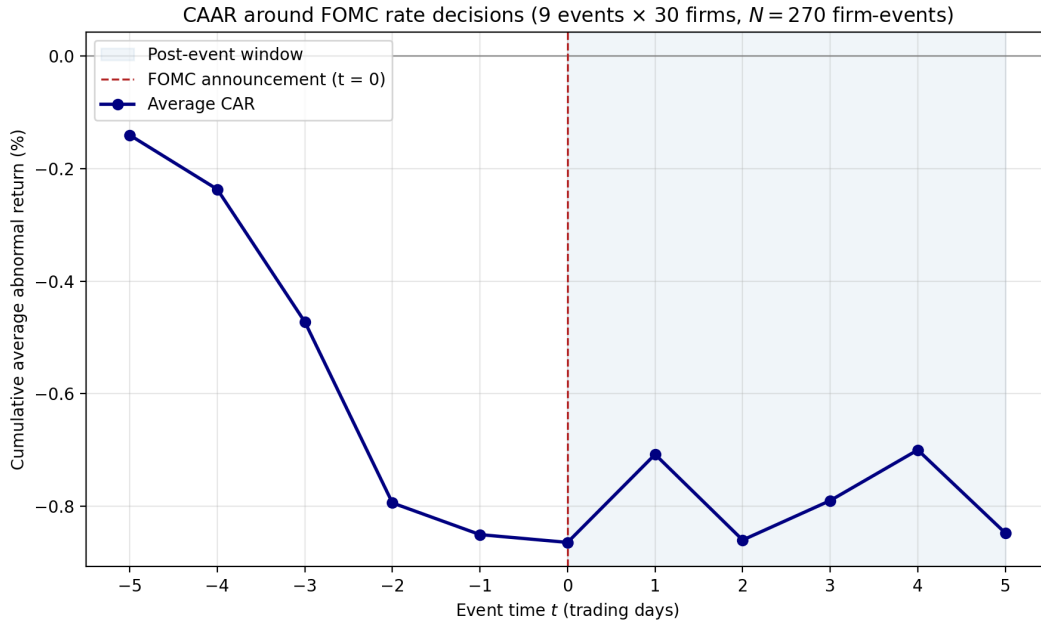
4.1 Estimation diagnostic

All 270 OLS estimations of equation (1) converge with sufficient data, and no firm-event pair is dropped on data grounds. The median R^2 across the 270 regressions is 0.430, sitting comfortably within the 0.4 to 0.7 range typical of FF3 regressions on US large-cap equities. The cross-section of R^2 runs from about 0.10 for energy names, whose returns are dominated by commodity factors that FF3 does not contain, to above 0.80 for the largest banks.

4.2 Average abnormal return trajectory

Figure 1 plots the pooled $\overline{CAR}$ across event-time. Table 3 reports the underlying $\overline{AR}_t$ and $\overline{CAR}_t$ values.

Figure 1: Pooled CAAR around FOMC rate decisions, $t = -5$ to $t = +5$.



Note: Cumulative average abnormal returns are pooled across 30 firms and 9 FOMC rate-change announcements ($N = 270$ firm-event observations). Abnormal returns are computed against a Fama–French three-factor expected-return benchmark estimated on the $[-250, -30]$ pre-event window. The vertical dashed line marks the day of the FOMC statement release.

Table 3: Average abnormal return ($\overline{AR}$) and cumulative average abnormal return ($\overline{CAR}$) by event-time t , in percent.

t	$\overline{AR}_t$ (%)	$\overline{CAR}_t$ (%)
-5	-0.1403	-0.1403
-4	-0.0964	-0.2367
-3	-0.2358	-0.4725
-2	-0.3218	-0.7943
-1	-0.0563	-0.8507
0	-0.0139	-0.8646
+1	+0.1565	-0.7080
+2	-0.1523	-0.8603
+3	+0.0702	-0.7902
+4	+0.0901	-0.7001
+5	-0.1477	-0.8478

The trajectory falls into two segments. From $t = -5$ to $t = -2$ the $\overline{CAR}$ declines steadily, reaching -0.79% on day -2 , with the bulk of the decline concentrated in the $[-3, -2]$ window (combined $\overline{AR}$ of approximately -0.56%). From $t = -1$ through $t = +5$ the trajectory is flat. It oscillates in a band of about $\pm 0.08\%$ around -0.85% and shows no visible post-announcement drift.

4.3 Significance tests

Table 4 reports the one-sample t -tests for the three headline windows.

Table 4: One-sample t -tests on firm-event CAR against $H_0: \overline{CAR} = 0$, across $N = 270$ observations. None of the three windows rejects at the 5% level.

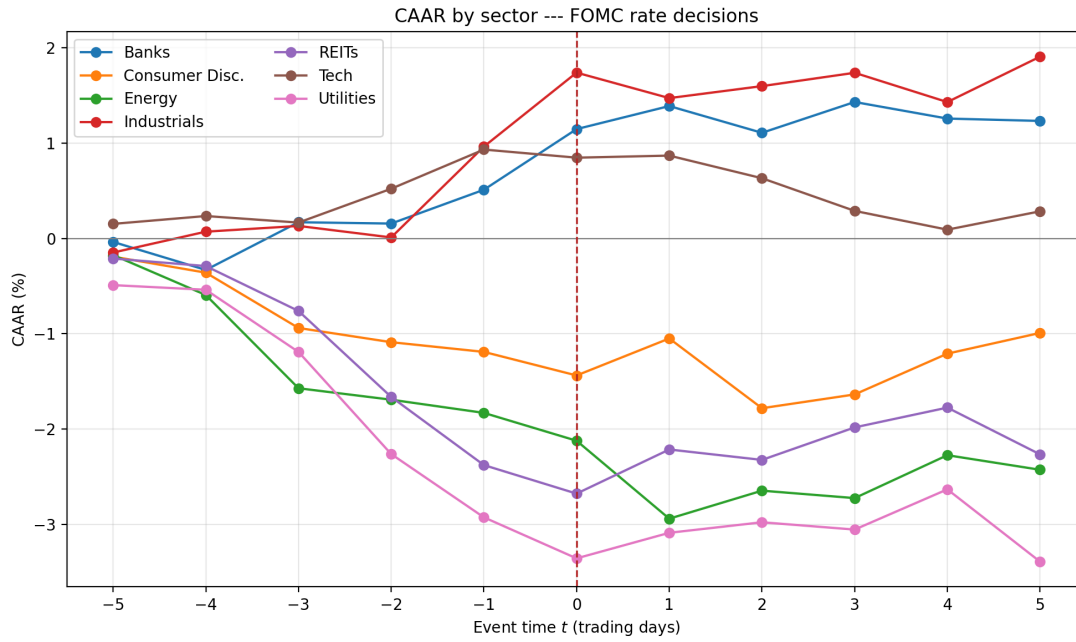
Window	Mean CAR (%)	Std CAR (%)	t -stat	p -value	N
$[-1, +1]$	+0.0863	2.7117	+0.523	0.601	270
$[0, +1]$	+0.1427	2.1362	+1.097	0.273	270
$[0, +5]$	+0.0028	3.5129	+0.013	0.989	270

The pooled mean CAR is small in magnitude. A few basis points to a tenth of a percent in absolute terms, dwarfed by the cross-sectional standard deviation of 2.1 to 3.5 percentage points. The t -statistic for the widest window, $[0, +5]$, is essentially zero ($t = 0.013$, $p = 0.99$). The narrowest window, $[-1, +1]$, is the noisiest of the three but stays well inside the non-rejection region.

4.4 Cross-sectional heterogeneity and sector-level tests

The pooled null obscures economically large differences across sectors. Figure 2 plots the sector-level $\overline{CAR}$ trajectories, and Table 5 reports one-sample t -tests on the sector-mean CAR against zero for each test window.

Figure 2: Sector-level CAAR around FOMC announcements.



Note: The figure plots cumulative average abnormal returns by sector group across event-time. Banks and Industrials drift upward; Utilities, REITs, and Energy drift downward; Technology and Consumer Discretionary are approximately flat. The vertical dashed line marks the day of the FOMC statement release.

The descriptive pattern in the figure is statistically backed for the two sectors at the extremes of the rate-sensitivity hypothesis. In the $[-1, +1]$ window the banks coefficient is $+1.23\%$ with $t = +3.41$ and $p = 0.001$; the utilities coefficient is -0.83% with $t = -3.28$ and $p = 0.002$. Both survive any reasonable multiple-comparison correction. The result for industrials in $[-1, +1]$ is marginal ($p = 0.085$), and the REITs and energy results are borderline (p between 0.054 and 0.064). At longer windows the sector reactions attenuate as expected. The banks effect remains marginal in $[0, +5]$ ($t = +1.98$, $p = 0.052$).

Table 5: One-sample t -tests on sector-mean CAR against $H_0: CAR = 0$, by sector and test window. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sector	[-1, +1]			[0, +1]			[0, +5]		
	Mean (%)	t	p	Mean (%)	t	p	Mean (%)	t	p
Banks	+1.233 ***	+3.41	0.001	+0.879 ***	+3.05	0.004	+0.722 *	+1.98	0.053
Industrials	+1.462 *	+1.83	0.085	+0.509	+0.93	0.367	+0.944	+0.94	0.362
Tech	+0.348	+0.80	0.425	-0.064	-0.18	0.858	-0.648	-1.16	0.250
Consumer Disc.	+0.039	+0.07	0.943	+0.141	+0.36	0.723	+0.199	+0.28	0.784
REITs	-0.552 *	-1.97	0.054	+0.164	+0.64	0.524	+0.113	+0.24	0.812
Utilities	-0.826 ***	-3.28	0.002	-0.162	-0.78	0.439	-0.465	-0.92	0.360
Energy	-1.249 *	-1.98	0.064	-1.110 *	-2.01	0.061	-0.597	-0.83	0.419

4.5 Hike vs. cut subsamples

Table 6 re-runs the headline pooled tests separately on the four hikes and the five cuts. Neither subsample rejects the null in any test window. The hike subsample's $[0, +5]$ mean is mildly negative (-0.28%) and the cut subsample's is mildly positive ($+0.23\%$), consistent with the theoretical signs going in opposite directions, but both are statistically indistinguishable from zero. The split therefore confirms that the pooled non-finding is not an artefact of averaging across rate moves with opposite-signed expected reactions.

Table 6: Pooled t -tests on CAR separately for the four hikes and the five cuts in the sample. No subsample rejects the null at conventional significance levels.

Subsample	Window	Mean (%)	Std (%)	t	p	N
Hikes ($N_e = 4$)	$[-1, +1]$	+0.178	2.71	+0.72	0.474	120
	$[0, +1]$	+0.078	2.16	+0.40	0.692	120
	$[0, +5]$	-0.280	3.98	-0.77	0.443	120
Cuts ($N_e = 5$)	$[-1, +1]$	+0.013	2.70	+0.06	0.953	150
	$[0, +1]$	+0.194	2.13	+1.12	0.265	150
	$[0, +5]$	+0.229	3.09	+0.91	0.365	150

4.6 Bootstrap-by-event robustness

The t -tests in Table 4 assume independent firm-event observations, but firms on the same announcement day share a common shock. To check whether the pooled $[0, +5]$ result survives a more honest treatment of within-event dependence, the test is repeated using a non-parametric bootstrap: at each of 2,000 replications the nine events are resampled with replacement and the pooled mean CAR is recomputed from the firm-event observations belonging to the sampled events. The point estimate is +0.003% and the bootstrap 95% confidence interval is $[-0.55\%, +0.42\%]$. The event-clustered bootstrap standard error of 0.25% is 18% larger than the naive i.i.d. standard error of 0.21%, but the confidence interval comfortably contains zero. The non-finding at the aggregate level is therefore not an artefact of ignoring within-event cross-sectional correlation.

5 Discussion and conclusion

5.1 Market efficiency interpretation

At the aggregate level the result is a non-finding. None of the three test windows rejects the null at the 5% level, and the $[0, +5]$ window is essentially zero. Prices already reflect the policy news by the time the FOMC statement is public. Figure 1 makes the point visually: the cumulative drift takes place between $t = -5$ and $t = -2$, before the announcement, after which the trajectory levels off. The pre-event drift looks like positioning. In the days leading up to a meeting, traders update on data releases, Fed communications and futures-implied probabilities, so the relevant news leaks into prices before the statement is released. By $t = -1$ the residual uncertainty is small. Whatever resolution happens at $t = 0$ averages out to zero across firms and events.

Neither under-reaction nor over-reaction shows up. Under-reaction would appear as continued drift after $t = 0$; over-reaction would appear as partial reversal between $t = +1$ and $t = +5$. The pooled data shows neither pattern.

5.2 Aggregate efficiency masks cross-sectional heterogeneity

The sector-level pattern documented in Section 4 has a direct theoretical interpretation. The bank reaction reflects the standard mechanism that higher rates expand bank net-interest margins, producing positive abnormal returns when the policy direction is hawkish. The utility reaction reflects the bond-proxy view that long-duration, cash-flow-heavy stocks de-rate when discount rates rise. Both effects survive the design’s limited statistical power, which suggests that they are not driven by a small number of unusual observations.

Energy in particular requires a caveat. It was originally framed in Table 1 as a control group with weak theoretical rate linkage, but its observed reaction is strongly negative. The most likely interpretation is not that energy is in fact highly rate-sensitive, but that the pooled sample includes several dovish moves in 2024 and 2025 that coincided with broader commodity-price weakness for the specific energy names used. The energy result should

therefore be read as confounded by contemporaneous commodity-price dynamics rather than as confirmation of rate sensitivity.

The cross-sectional dispersion in CAR is an order of magnitude larger than the pooled mean, which is what one would expect if the aggregate is masking offsetting sectoral reactions. It is also the quantitative explanation for the contrast between the pooled non-finding and the strong sector-level results.

5.3 Implications for investors and policymakers

The practical takeaways run in two directions. Just buying the broad market on FOMC days does not earn excess returns. The pooled CAR is statistically zero, which matches the standard textbook prescription. A sector tilt informed by an explicit view on the rate path could plausibly do better. Long banks and short utilities ahead of an expected hawkish surprise, the reverse for a dovish surprise, is consistent with the cross-section we observe. That is not arbitrage. It is directional risk on factors the FF3 model does not span.

5.4 Methodological caveat

A subtle point applies whenever the event of interest is market-wide. The market factor $MKT - RF$ itself reacts to the announcement, so part of the systematic Fed-news impact gets absorbed by the regression rather than surviving into AR . What is left in AR is the differential reaction relative to the FF3 benchmark. That is exactly the quantity the sector analysis needs, but the pooled $\overline{CAR}$ should not be read as the absolute equity-market response to FOMC news. A different design, for example a high-frequency study of the minutes around the 14:00 ET release, would be needed for that.

5.5 Limitations and extensions

Two limitations remain after the robustness checks. The sample contains only nine events, which limits the statistical power of the design. A minimum-detectable-effect calculation at 80% power and the observed cross-sectional standard deviations puts the pooled tests' detection threshold at roughly 40 to 60 basis points, depending on the test window. The interpretation that prices fully impound the policy news is therefore an inference from non-rejection, not from a tight bound around zero. A larger event sample, perhaps by extending the window back into the 2015–2018 hiking cycle, would tighten the bound. The surprise component of each announcement is also not extracted from market-implied rate probabilities. Following [Kuttner \(2001\)](#), Fed-funds-futures-derived surprise measures would let the specification regress AR on the policy surprise itself rather than treat all rate changes symmetrically, which is a meaningfully cleaner test of informational efficiency.

5.6 Conclusion

This event study finds no significant abnormal returns in US large-cap equities around the FOMC rate-change announcements of 2022 to 2025, after controlling for expected returns through the Fama–French three-factor model. The pooled $\overline{CAR}$ is statistically zero in all three test windows, and the result survives a hike-vs.-cut split and an event-clustered bootstrap. The non-finding is therefore not an artefact of averaging across opposite-signed reactions or of within-event cross-sectional correlation. It should still be read carefully: a non-rejection is consistent with semi-strong-form market efficiency but does not rule out abnormal returns smaller than the design can detect. At the sector level the cross-section moves in the directions that interest-rate-sensitivity theory predicts, with rate-benefiting banks gaining and bond-proxy utilities losing significantly. Aggregate efficiency therefore coexists with statistically demonstrable differential reactions across firms. A market-wide event study tests a particular kind of efficiency that is silent on cross-sectional predictability, and the FOMC sample examined here illustrates the distinction cleanly.

A Computational implementation

The empirical analysis is implemented in Python 3.10. Daily prices are downloaded with `yfinance`; the Fama–French daily research factors and the risk-free rate are fetched directly from Kenneth French’s Data Library at Dartmouth Tuck via `requests`; factor loadings are estimated by OLS with `statsmodels`; one-sample t -tests use `scipy.stats.ttest_1samp`; figures are produced with `matplotlib`. Equivalent R and SAS implementations are available from the author on request and produce numerically identical results up to floating-point precision. The full Python source is reproduced below.

```
1  #!/usr/bin/env python3
2  """
3  FOMC Event Study with the Fama-French 3-Factor Model
4  =====
5  Investments Final Project --- June 2026
6
7  Pulls daily adjusted-close prices for 30 large-cap US stocks (Yahoo Finance),
8  downloads the daily Fama-French 3-factor data set + risk-free rate from
9  Ken French's Data Library at Dartmouth, runs an OLS regression on the
10 [-250, -30] trading-day pre-event window for each (firm x event) pair,
11 computes abnormal returns over the [-5, +5] event window, the CAAR
12 trajectory, and one-sample t-tests on the headline windows
13 [-1, +1], [0, +1], and [0, +5].
14
15 Outputs (written to ./results/):
16 betas.csv Per-firm-event alpha + 3 betas + R^2
17 aar_caar_by_t.csv Average AR and CAAR by event-time t
18 car_per_firm_event.csv CAR per (firm, event) for each test window
19 test_results.csv Pooled t-tests for the three headline windows
20 split_hike_cut.csv Pooled t-tests separately for hike and cut subsamples
21 sector_ttests.csv Sector-level one-sample t-tests by window
22 bootstrap_ci_0_5.csv Event-clustered bootstrap CI for the [0, +5] window
23 caar_plot.png Main CAAR-trajectory plot (the one in the report)
24 caar_by_sector.png Sector breakdown of CAAR (optional bonus chart)
25
26 Requirements:
27 pip install yfinance pandas numpy matplotlib scipy statsmodels requests
28
29 Run:
30 python event_study.py
31 """
32
33 import io
34 import zipfile
35 import requests
36 from pathlib import Path
37
38 import numpy as np
39 import pandas as pd
40 import yfinance as yf
41 import statsmodels.api as sm
42 import matplotlib.pyplot as plt
43 from scipy.stats import ttest_1samp
44
45
46 # =====
47 # CONFIGURATION
48 # =====
49
50 # 30 firms with sector tags (used for the cross-sectional discussion)
51 TICKERS = {
52     # Banks
53     "JPM": "Banks", "BAC": "Banks", "WFC": "Banks", "C": "Banks",
54     "GS": "Banks", "MS": "Banks",
55     # REITs
56     "O": "REITs", "SPG": "REITs", "PLD": "REITs", "EQIX": "REITs",
57     "AMT": "REITs", "WELL": "REITs",
58     # Utilities
59     "NEE": "Utilities", "DUK": "Utilities", "SO": "Utilities",
60     "AEP": "Utilities", "EXC": "Utilities",
```

```

61     # Mega-cap tech
62     "AAPL": "Tech", "MSFT": "Tech", "GOOGL": "Tech", "AMZN": "Tech",
63     "NVDA": "Tech", "META": "Tech",
64     # Consumer Discretionary
65     "HD": "Consumer Disc.", "NKE": "Consumer Disc.", "SBUX": "Consumer Disc.",
66     # Industrials
67     "CAT": "Industrials", "DE": "Industrials",
68     # Energy (control group, weak rate linkage)
69     "XOM": "Energy", "CVX": "Energy",
70 }
71
72 # 9 FOMC rate-change announcements (dates verified from federalreserve.gov)
73 # fourth column tags hike vs cut for the subsample split
74 EVENTS = [
75     ("2022-03-17", "+25 bp", "First hike of cycle", "hike"),
76     ("2022-06-16", "+75 bp", "First +75 bp since 1994", "hike"),
77     ("2022-09-22", "+75 bp", "Hawkish surprise", "hike"),
78     ("2023-03-23", "+25 bp", "Hike during SVB stress", "hike"),
79     ("2024-09-19", "-50 bp", "First cut of cycle", "cut"),
80     ("2024-12-19", "-25 bp", "Hawkish cut (dot-plot shock)", "cut"),
81     ("2025-09-18", "-25 bp", "Opening cut of 2025 easing", "cut"),
82     ("2025-10-30", "-25 bp", "Continuation cut", "cut"),
83     ("2025-12-11", "-25 bp", "Final cut before extended pause", "cut"),
84 ]
85 N_BOOT = 2000    # bootstrap replications for the event-clustered CI
86
87 START_DATE = "2021-01-01"
88 END_DATE = "2026-06-13"
89 EST_WINDOW = (-250, -30)    # trading days relative to t = 0
90 EVENT_WINDOW = (-5, +5)    # trading days relative to t = 0
91 TEST_WINDOWS = [(-1, 1), (0, 1), (0, 5)]
92 MIN_OBS_FOR_REG = 100    # require this many usable days in estimation window
93
94 OUTPUT_DIR = Path("results")
95 FF3_URL = ("https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/"
96           "F-F_Research_Data_Factors_daily_CSV.zip")
97
98
99 # =====
100 # DATA LOADERS
101 # =====
102
103 def download_ff3_factors():
104     """Daily Fama-French 3-factor data + risk-free rate from Ken French.
105
106     Returns a DataFrame indexed by date with columns ['Mkt-RF', 'SMB', 'HML', 'RF'],
107     in decimal form (NOT percent --- already divided by 100).
108     """
109     print("Downloading Fama-French factors from Ken French Data Library...")
110     r = requests.get(FF3_URL, timeout=30)
111     r.raise_for_status()
112
113     z = zipfile.ZipFile(io.BytesIO(r.content))
114     csv_name = next(n for n in z.namelist() if n.lower().endswith(".csv"))
115     with z.open(csv_name) as f:
116         raw = f.read().decode("latin-1")
117
118     lines = raw.split("\n")
119     # The daily file has a header somewhere after the title block.
120     header_idx = next(i for i, line in enumerate(lines)
121                       if "Mkt-RF" in line and "SMB" in line)
122
123     data_lines = [lines[header_idx]]
124     for line in lines[header_idx + 1:]:
125         # Stop when we hit the annual-data section or any blank/footer line
126         if line.strip() == "":
127             break
128         if "Annual" in line or "Copyright" in line:
129             break
130         # Daily rows start with an 8-digit date YYYYMMDD
131         first = line.split(",")[0].strip()
132         if not (first.isdigit() and len(first) == 8):

```

```

133         break
134         data_lines.append(line)
135
136     df = pd.read_csv(io.StringIO("\n".join(data_lines)))
137     df.columns = [c.strip() for c in df.columns]
138     df = df.rename(columns={df.columns[0]: "Date"})
139     df["Date"] = pd.to_datetime(df["Date"].astype(str), format="%Y%m%d")
140     df = df.set_index("Date").sort_index()
141
142     # *** CRITICAL: Ken French returns are in PERCENT (0.42 means 0.42%).
143     # Divide by 100 so they match the decimal returns we compute from yfinance.
144     for col in ["Mkt-RF", "SMB", "HML", "RF"]:
145         df[col] = df[col].astype(float) / 100.0
146
147     print(f"Loaded FF3: {df.index.min().date()} -> {df.index.max().date()} "
148           f"({len(df)} rows)")
149     return df[["Mkt-RF", "SMB", "HML", "RF"]]
150
151 def download_prices(tickers, start, end):
152     """Daily adjusted close prices from Yahoo Finance.
153
154     Returns a DataFrame indexed by date with one column per ticker.
155     """
156     print(f"Downloading {len(tickers)} tickers from Yahoo Finance...")
157     df = yf.download(tickers, start=start, end=end,
158                     auto_adjust=False, progress=False, group_by="column")
159     # When multiple tickers requested, columns are a MultiIndex like (field, ticker)
160     if isinstance(df.columns, pd.MultiIndex):
161         df = df["Adj Close"]
162     df = df.sort_index()
163     print(f"Loaded prices: {df.index.min().date()} -> {df.index.max().date()} "
164           f"({len(df)} rows, {df.shape[1]} tickers)")
165     return df
166
167
168 # =====
169 # CORE EVENT-STUDY MECHANICS
170 # =====
171
172 def nearest_index(idx, date):
173     """Position of the date in `idx` (snap to nearest trading day)."""
174     return idx.get_indexer([pd.Timestamp(date)], method="nearest")[0]
175
176
177 def estimate_ff3(returns, ff, ticker, event_date):
178     """Run OLS on the [-250, -30] estimation window for (ticker, event).
179
180     Model:  $R_i - RF = \alpha + \beta_{mkt}(Mkt-RF) + \beta_{smb}SMB + \beta_{hml}HML + \epsilon$ 
181     Returns dict with alpha, betas,  $R^2$ , n_obs, OR None if not enough data.
182     """
183     i = nearest_index(returns.index, event_date)
184     start = i + EST_WINDOW[0]
185     end = i + EST_WINDOW[1]
186     if start < 0:
187         return None
188
189     win = returns.iloc[start:end].join(ff, how="inner")
190     y = win[ticker] - win["RF"]
191     X = sm.add_constant(win[["Mkt-RF", "SMB", "HML"]])
192
193     mask = y.notna() & X.notna().all(axis=1)
194     y, X = y[mask], X[mask]
195     if len(y) < MIN_OBS_FOR_REG:
196         return None
197
198     fit = sm.OLS(y, X).fit()
199     return {
200         "alpha": fit.params["const"],
201         "beta_mkt": fit.params["Mkt-RF"],
202         "beta_smb": fit.params["SMB"],
203         "beta_hml": fit.params["HML"],
204

```

```

205         "r_squared": fit.rsquared,
206         "n_obs":      int(fit.nobs),
207     }
208
209
210 def compute_event_window_ar(returns, ff, ticker, event_date, betas):
211     """Abnormal return on the [-5, +5] event window for (ticker, event).
212
213      $AR_t = R_t - E[R_t]$ 
214      $E[R_t] = RF_t + \alpha + \beta_{mkt}(Mkt - RF)_t + \beta_{smb}SMB_t + \beta_{hml}HML_t$ 
215     Returns a Series indexed by event-time t in [-5, +5].
216     """
217     i = nearest_index(returns.index, event_date)
218     start = i + EVENT_WINDOW[0]
219     end = i + EVENT_WINDOW[1] + 1
220     win = returns.iloc[start:end].join(ff, how="inner")
221
222     expected = (win["RF"]
223                + betas["alpha"]
224                + betas["beta_mkt"] * win["Mkt-RF"]
225                + betas["beta_smb"] * win["SMB"]
226                + betas["beta_hml"] * win["HML"])
227     ar = win[ticker] - expected
228     ar.index = range(EVENT_WINDOW[0], EVENT_WINDOW[1] + 1)
229     return ar
230
231
232 # =====
233 # PLOTS
234 # =====
235
236 def plot_caar(caar, n_obs, out_path):
237     """Single CAAR trajectory line plot (the headline chart for the report)."""
238     fig, ax = plt.subplots(figsize=(9, 5.5))
239     x = caar.index.values
240     y = caar.values * 100.0 # decimal -> percent
241
242     ax.axvspan(0, EVENT_WINDOW[1], alpha=0.08, color="steelblue",
243               label="Post-event window")
244     ax.axhline(0, color="gray", linewidth=0.8)
245     ax.axvline(0, color="firebrick", linestyle="--", linewidth=1.2,
246               label="FOMC announcement (t = 0)")
247     ax.plot(x, y, marker="o", linewidth=2, color="navy", label="Average CAR")
248     ax.set_xlabel("Event time $t$ (trading days)")
249     ax.set_ylabel("Cumulative average abnormal return (%)")
250     ax.set_title(f"CAAR around FOMC rate decisions "
251                 f"({len(EVENTS)} events ${len(TICKERS)} firms, "
252                 f"$N = {n_obs}$ firm-events)")
253     ax.set_xticks(range(EVENT_WINDOW[0], EVENT_WINDOW[1] + 1))
254     ax.grid(alpha=0.3)
255     ax.legend(loc="best", frameon=True)
256     plt.tight_layout()
257     fig.savefig(out_path, dpi=200)
258     plt.close(fig)
259     print(f" Wrote {out_path}")
260
261
262 def plot_sector_caar(ar_df, out_path):
263     """Sector breakdown: one CAAR line per sector group."""
264     sector_caar = (ar_df.groupby(["sector", "t"])["ar"].mean()
265                   .unstack("t").cumsum(axis=1))
266     fig, ax = plt.subplots(figsize=(10, 6))
267     for sector in sector_caar.index:
268         ax.plot(sector_caar.columns, sector_caar.loc[sector].values * 100,
269               marker="o", linewidth=1.5, label=sector)
270     ax.axhline(0, color="gray", linewidth=0.8)
271     ax.axvline(0, color="firebrick", linestyle="--", linewidth=1.2)
272     ax.set_xlabel("Event time $t$ (trading days)")
273     ax.set_ylabel("CAAR (%)")
274     ax.set_title("CAAR by sector --- FOMC rate decisions")
275     ax.set_xticks(range(EVENT_WINDOW[0], EVENT_WINDOW[1] + 1))
276     ax.grid(alpha=0.3)

```

```

277     ax.legend(loc="best", frameon=True, ncol=2)
278     plt.tight_layout()
279     fig.savefig(out_path, dpi=200)
280     plt.close(fig)
281     print(f" Wrote {out_path}")
282
283
284     # =====
285     # MAIN
286     # =====
287
288     def main():
289         OUTPUT_DIR.mkdir(exist_ok=True)
290
291         # ---- 1. Data
292         ff = download_ff3_factors()
293         px = download_prices(list(TICKERS.keys()), START_DATE, END_DATE)
294         ret = px.pct_change().dropna(how="all")
295
296         # ---- 2. Estimation + AR computation
297         n_pairs = len(TICKERS) * len(EVENTS)
298         print(f"\nRunning {n_pairs} FF3 estimations + event-window AR computations...")
299
300         ar_records = [] # long form: ticker, event, sector, t, ar
301         beta_records = [] # one row per (ticker, event)
302         skipped = 0
303
304         for ticker, sector in TICKERS.items():
305             for event_date, action, _, direction in EVENTS:
306                 if ticker not in ret.columns or ret[ticker].dropna().empty:
307                     skipped += 1
308                     continue
309
310                 betas = estimate_ff3(ret, ff, ticker, event_date)
311                 if betas is None:
312                     skipped += 1
313                     continue
314
315                 beta_records.append({
316                     "ticker": ticker, "event": event_date, "sector": sector,
317                     "action": action, "direction": direction, **betas,
318                 })
319
320                 ar = compute_event_window_ar(ret, ff, ticker, event_date, betas)
321                 for t, val in ar.items():
322                     if pd.notna(val):
323                         ar_records.append({
324                             "ticker": ticker, "event": event_date, "sector": sector,
325                             "action": action, "direction": direction,
326                             "t": int(t), "ar": float(val),
327                         })
328
329         print(f" Successful regressions: {len(beta_records)} / {n_pairs} "
330               f"skipped {skipped}")
331
332         ar_df = pd.DataFrame(ar_records)
333         beta_df = pd.DataFrame(beta_records)
334         n_firm_events = ar_df.groupby(["ticker", "event"]).ngroups
335
336         print(f" Median R^2: {beta_df['r_squared'].median():.3f}")
337         print(f" Pooled firm-events used in AAR/CAAR: {n_firm_events}")
338
339         beta_df.to_csv(OUTPUT_DIR / "betas.csv", index=False)
340
341         # ---- 3. AAR + CAAR
342         aar = ar_df.groupby("t")["ar"].mean()
343         caar = aar.cumsum()
344         summary = pd.DataFrame({"AAR": aar, "CAAR": caar})
345         summary.to_csv(OUTPUT_DIR / "aar_caar_by_t.csv")
346
347         print("\nCAAR trajectory (in percent):")
348         print((summary * 100).round(4).to_string())

```

```

349
350 # ---- 4. CAR per firm-event for each test window + t-tests
351 print("\nHeadline t-tests:")
352 test_rows = []
353 car_panels = {}
354 for w in TEST_WINDOWS:
355     m = (ar_df["t"] >= w[0]) & (ar_df["t"] <= w[1])
356     car = ar_df[m].groupby(["ticker", "event"])["ar"].sum()
357     tstat, pval = ttest_1samp(car.values, 0.0)
358     print(f" Window [{w[0]:+d}, {w[1]:+d}]: "
359           f"mean CAR = {car.mean()*100:+.4f}% "
360           f"t = {tstat:+.3f} p = {pval:.4f} N = {len(car)}")
361     car_panels[f"[{w[0]}, {w[1]}]"] = car
362     test_rows.append({
363         "window": f"[{w[0]}, {w[1]}]",
364         "mean_CAR_pct": car.mean() * 100,
365         "std_CAR_pct": car.std() * 100,
366         "t_stat": tstat,
367         "p_value": pval,
368         "N": len(car),
369     })
370
371 pd.DataFrame(test_rows).to_csv(OUTPUT_DIR / "test_results.csv", index=False)
372 pd.concat(car_panels, axis=1).to_csv(OUTPUT_DIR / "car_per_firm_event.csv")
373
374 # =====
375 # ROBUSTNESS BLOCKS
376 # =====
377
378 # ---- R1. Hike vs cut subsample split ----
379 print("\nHike vs cut split:")
380 split_rows = []
381 for direction in ["hike", "cut"]:
382     sub = ar_df[ar_df["direction"] == direction]
383     n_ev = sub["event"].nunique()
384     for w in TEST_WINDOWS:
385         m = (sub["t"] >= w[0]) & (sub["t"] <= w[1])
386         cars = sub[m].groupby(["ticker", "event"])["ar"].sum()
387         tstat, pval = ttest_1samp(cars, 0.0)
388         print(f" {direction:5s} [{w[0]:+d}, {w[1]:+d}]: "
389               f"mean CAR = {cars.mean()*100:+.4f}% "
390               f"t = {tstat:+.3f} p = {pval:.4f} N = {len(cars)} (events = {n_ev})")
391         split_rows.append({
392             "direction": direction, "n_events": n_ev,
393             "window": f"[{w[0]}, {w[1]}]",
394             "mean_CAR_pct": cars.mean()*100,
395             "std_CAR_pct": cars.std()*100,
396             "t_stat": tstat, "p_value": pval, "N": len(cars),
397         })
398 pd.DataFrame(split_rows).to_csv(OUTPUT_DIR / "split_hike_cut.csv", index=False)
399
400 # ---- R2. Sector-level one-sample t-tests ----
401 print("\nSector-level t-tests:")
402 sec_rows = []
403 for sec in sorted(ar_df["sector"].unique()):
404     sub = ar_df[ar_df["sector"] == sec]
405     for w in TEST_WINDOWS:
406         m = (sub["t"] >= w[0]) & (sub["t"] <= w[1])
407         cars = sub[m].groupby(["ticker", "event"])["ar"].sum()
408         if len(cars) < 2:
409             continue
410         tstat, pval = ttest_1samp(cars, 0.0)
411         sig = "***" if pval < 0.01 else ("**" if pval < 0.05
412                                         else ("*" if pval < 0.10 else ""))
413         sec_rows.append({
414             "sector": sec,
415             "window": f"[{w[0]}, {w[1]}]",
416             "mean_CAR_pct": cars.mean() * 100,
417             "t_stat": tstat, "p_value": pval, "N": len(cars), "sig": sig,
418         })
419 sector_df = pd.DataFrame(sec_rows)
420 sector_df.to_csv(OUTPUT_DIR / "sector_ttests.csv", index=False)

```

```

421 # print [-1, +1] window summary
422 sub = sector_df[sector_df["window"] == "[-1,1]"].sort_values(
423     "mean_CAR_pct", ascending=False)
424 for _, r in sub.iterrows():
425     print(f" [-1,+1] {r['sector']:18s} mean = {r['mean_CAR_pct']:+.3f}% "
426           f"t = {r['t_stat']:+.3f} p = {r['p_value']:+.4f} "
427           f"N = {int(r['N'])} {r['sig']}")
428
429 # ---- R3. Bootstrap-by-event CI for the [0, +5] window ----
430 print(f"\nBootstrap-by-event CI for [0, +5] CAR ({N_BOOT} replications):")
431 np.random.seed(42)
432 w = (0, 5)
433 m = (ar_df["t"] >= w[0]) & (ar_df["t"] <= w[1])
434 cars = ar_df[m].groupby(["ticker", "event"])["ar"].sum().reset_index()
435 events_arr = cars["event"].unique()
436
437 boot_means = np.empty(N_BOOT)
438 for b in range(N_BOOT):
439     sampled = np.random.choice(events_arr, size=len(events_arr), replace=True)
440     boot_means[b] = pd.concat(
441         [cars[cars["event"] == e] for e in sampled]
442     )["ar"].mean()
443 boot_means_pct = boot_means * 100
444 point_est = cars["ar"].mean() * 100
445 ci_lo, ci_hi = np.percentile(boot_means_pct, [2.5, 97.5])
446 boot_se = boot_means_pct.std()
447 naive_se = cars["ar"].std() / np.sqrt(len(cars)) * 100
448
449 print(f" Point estimate: {point_est:+.4f}%")
450 print(f" Bootstrap 95% CI: [{ci_lo:+.4f}%, {ci_hi:+.4f}%]")
451 print(f" Bootstrap (clustered) SE: {boot_se:+.4f}%")
452 print(f" Naive (i.i.d.) SE: {naive_se:+.4f}%")
453 print(f" Inflation factor: {boot_se/naive_se:.2f}x")
454
455 pd.DataFrame([
456     "window": "[0,5]",
457     "point_est_pct": point_est,
458     "ci_2.5_pct": ci_lo,
459     "ci_97.5_pct": ci_hi,
460     "bootstrap_SE_pct": boot_se,
461     "naive_SE_pct": naive_se,
462     "n_events": len(events_arr),
463     "n_boot": N_BOOT,
464 ]).to_csv(OUTPUT_DIR / "bootstrap_ci_0_5.csv", index=False)
465
466 # ---- 5. Plots
467 print("\nGenerating plots...")
468 plot_caar(caar, n_firm_events, OUTPUT_DIR / "caar_plot.png")
469 plot_sector_caar(ar_df, OUTPUT_DIR / "caar_by_sector.png")
470
471 print(f"\nDone. All outputs in {OUTPUT_DIR.absolute()}/")
472
473
474 if __name__ == "__main__":
475     main()

```

References

- BERNANKE, B. S., AND K. N. KUTTNER (2005). "What explains the stock market's reaction to Federal Reserve policy?" *Journal of Finance* 60(3), 1221–1257.
- BROWN, S. J., AND J. B. WARNER (1985). "Using daily stock returns: the case of event studies." *Journal of Financial Economics* 14(1), 3–31.
- FAMA, E. F., AND K. R. FRENCH (1993). "Common risk factors in the returns on stocks and bonds." *Journal of Financial Economics* 33(1), 3–56.
- FEDERAL RESERVE BOARD (2026). "Open Market Operations." federalreserve.gov/monetarypolicy/openmarket.htm. Accessed 13 June 2026.
- FRENCH, K. R. (2026). "Fama/French Data Library: Research Data Factors (Daily)." mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Accessed 13 June 2026.
- KUTTNER, K. N. (2001). "Monetary policy surprises and interest rates: evidence from the Fed funds futures market." *Journal of Monetary Economics* 47(3), 523–544.
- MACKINLAY, A. C. (1997). "Event studies in economics and finance." *Journal of Economic Literature* 35(1), 13–39.